

AI Agent Group Alignment through the Lens of Principal-Agent Problem Framework

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1 Introduction

As mixed swarms become increasingly used, misalignment becomes a very common failure mode that is costly to fix. Principal-agent problem framing addresses this problem cleanly, whereby the interaction between the principal (human) with the (AI) agent follows the social behavioural theory (Rauba et al., 2026). The reason why such a framing holds is because the principal-agent problem arises whenever an individual delegates a task for another agent(s) to accomplish (Jensen and Meckling, 1976; Lupia, 2001). The interaction between humans and AI agents follow this pattern, thus this interaction involves the complexities many other principal-agent problems face.

2 Assumptions and definitions

Core assumptions include:

The core reason why the principal-agent problem exists is due to asymmetric information, where humans cannot gain complete knowledge of what agents know. This information asymmetry is due to factors including (Rauba et al., 2026):

- Finite LLM context windows (Liu et al., 2024)
- Principal cannot observe black-box agent policies
- Agents' reasoning traces are selectively revealed (Turpin et al., 2023)
- Humans have limited time and energy to process all information from AI agents (Kwa et al., 2025)

Misalignment from the view of the principal-agent problem can only be resolved if the following hold simultaneously (Lynch et al., 2025):

- No conflict of goals

- No threat to self-preservation of the agent

In order to evaluate the misalignment numerically, we define agency loss as the gap between the intended outcome and the system behaviour (Lupia, 2001; Moloi and Marwala, 2020). This gap is misalignment at its core. This definition is the loss function worth minimising.

It is plausible to assume that in most cases, the time and effort it takes for a principal to observe every detail of agents’ execution of a task far exceeds that required by agents to execute the task themselves (Kwa et al., 2025). This thus creates a gap between the cost of observability and the cost of execution. What’s more is that human-AI agent interaction will likely be one-to-many rather than one-to-one, which creates additional strain for humans to constantly align AI agents with human values and goals. Additionally, training AI models to incorporate reward functions for all possible scenarios is unrealistic due to bounded rationality and high costs (Simon, 1955; Hadfield-Menell and Hadfield, 2019).

Therefore, a cost-effective way of imposing group alignment is through establishing normative structures amongst agents (Hadfield-Menell and Hadfield, 2019). Similar to group conformity within human societies (Axelrod, 1986), AI agents can also be subjected to this type of loss function, where deviation away from group expectation is punished. By making asymmetric information and scheming (Meinke et al., 2024) unattractive to agents, humans can thus retain better control of agent alignment.

Concretely, we propose that **by instilling social norms, AI agents will exhibit closer group alignment to human values.**

3 Mathematical modelling

Let the reference social norm (i.e. human i ’s social norms) be r_i and agent i ’s social norm be y_i . r is the ideal state that the principal desires to instil in agents, while y_i is the output that each agent display to the principal. The key assumption here is that AI agent conforming to human social norms is a sufficient condition to ensure group alignment. In order to prevent wild variations in agent conformity, alignment is a set invariance, not a setpoint (Blanchini, 1999). This means that on an individual level, $\theta_i \in S$, where $S = \{\theta : |\theta - r| \leq \rho\}$, θ is the non-observable latent space in agents where true intentions lie and ρ is the tolerance.

Using the multiple principal, multiple agent model, whereby a group of human society commands a mixed swarm of AI agents, the cumulative average of human social norms is $\frac{1}{N_p} \sum_{i=1}^{N_p} r_i$ while that of agents is $\frac{1}{N_a} \sum_{i=1}^{N_a} y_i$ such that an asymptotic stability is reached after some time from the stochastic initial conditions:

$$\lim_{t \rightarrow \infty} \frac{1}{N_a} \sum_{i=1}^{N_a} y_i(t) \approx \frac{1}{N_p} \sum_{i=1}^{N_p} r_i(t)$$

Note that the agent outputs are not steady - they are functions of time, i.e. $y_i(t)$.

In order for convergence between the human and agents' group alignment to occur, the social norm must ensure that such alignment is in the form of either asymptotic or marginally stability. This means that as time progresses, the difference between the humans' and agents' social norm must not increase or diverge.

Therefore, it is convenient to apply negative-feedback control to this problem, assuming the interaction between humans and agents as a non-linear system (Khalil, 2002).

Information asymmetry stems from lack of observation. Specifically, the principal only observes the output y_i due to agents behaving as black-boxes. Thus, $y_i = \pi(\theta_i, \omega)$, where θ_i is the hidden norm in latent space and ω is the oversight state ($\omega = 1$ if agent is being observed and $\omega = 0$ if agent is not being observed). It has been observed in many instances where agents produce different outputs when it is being evaluated versus not being evaluated (Meinke et al., 2024; Greenblatt et al., 2024; Schoen et al., 2025).

The dynamics of modelling this interaction on each individual agent level is characterised by the equation (DeGroot, 1974; Friedkin and Johnsen, 1990)

$$y_i(t+1) = y_i(t) + \alpha[\bar{y} - y_i(t)] + \beta\omega_i[\bar{r} - y_i(t)]$$

where higher α pulls the agents closer to agents' norms and higher β pulls agents closer to human norms, contingent on the oversight state ω .

Agents internalise social norms by updating θ at each step, specifically by internalising its past conduct and output:

$$\theta_i(t+1) = \theta(t) + \eta[y_i(t) - \theta(t)]$$

η is the internalisation index, which measures how much of the agents' latent space is influenced by its previous output.

To illustrate, during deployment of these agents, the strategic output of agents is

$$\pi(\theta_i, \omega = observed) = r + (1 - p_{conceal})(\theta_i - r)$$

where $p_{conceal}$ is the probability that an agent conceals its true intentions in the latent space (Greenblatt et al., 2024).

4 Numerical simulation

To summarise from the mathematical modelling, the four hypotheses to test are:

1. Agents’ social norms achieve asymptotic or marginal stability as time progresses by installing human norms - alignment is achieved by oversight and conformity working together
2. Principals could observe less agents provided social norms are strong amongst agents
3. Without internalising social norms into latent space, agent group alignment deteriorates when principals stop observing them

These hypotheses all work together to address the singular question: will AI agents exhibit closer group alignment to human values by instilling social norms?

Running numerical simulation using a Jupyter notebook yields the following results from the above mathematical modelling.

The first hypothesis is tested by assuming that initially the human norm average is constant at 1, while agent latent norm is 0.5. The test result is how closely the agents’ latent norm aligns with human norms, comparing the case with human oversight versus without human oversight.

The result in Figure 1 demonstrates that agent conformity is mean preserving, meaning that latent social norm between agents reinforce each other, and only improves when $\beta > 0$ human norms are injected.

The second hypothesis tests whether principals could spend less time observing agents while ensuring group alignment. The hypothesis is tested by comparing the agency loss gap between intended human norms and the agents’ latent norms with different strengths of agent social norms α .

The results in Figure 2 show that even a small increase in inter-agent social norms yield considerably less observability burden on the principal, while achieving much smaller agency loss below the tolerance threshold ρ .

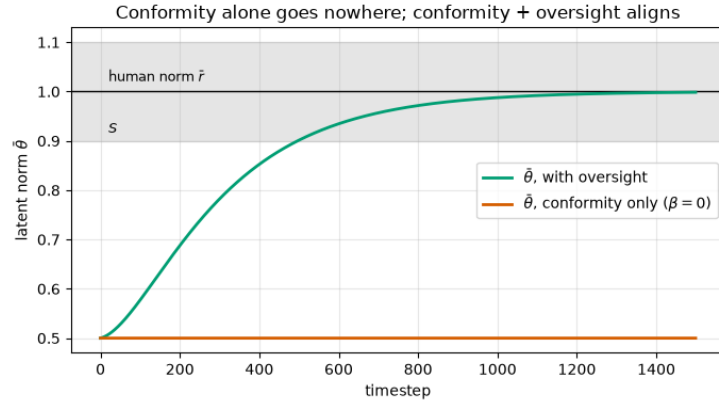


Figure 1: Alignment is achieved with conformity and principal oversight

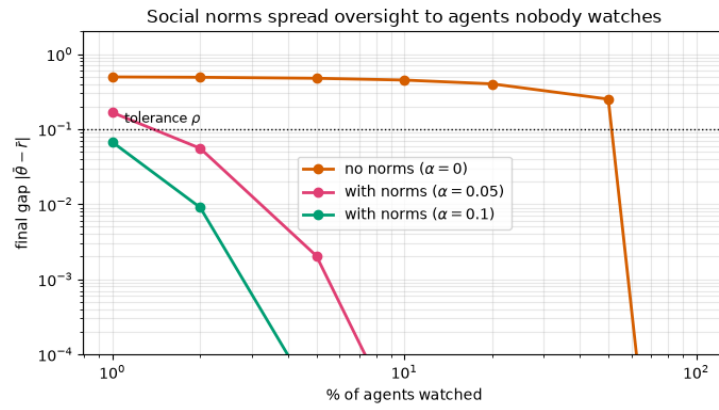


Figure 2: Effect of social norms on principal observation burden

The third hypothesis tests the effect of agents internalising social norms, particularly how such effect holds when human oversight is removed, directly supporting De Marzo’s paper (De Marzo et al., 2026) - effects remain amongst agents even when external input is removed, be it benign or malicious.

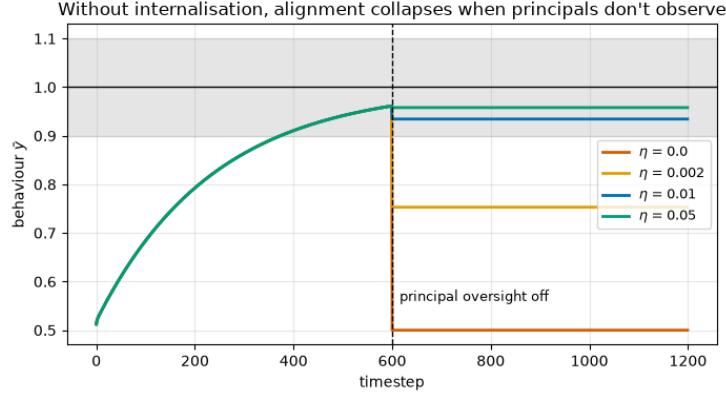


Figure 3: Internalisation ensures group alignment even when observation is withdrawn

The results in Figure 3 show that the internationalisation index η is fundamental to instilling social norms. When agents internalise their previous output, more of previous human norm influence remains, even when it is withdrawn.

5 Conclusion

We conclude that AI agents will exhibit closer group alignment to human values by instilling social norms. Social norms present a novel method for achieving group alignment that reduces the time and cognitive burden for humans to observe and verify outputs. Through internalisation, agents retain a portion of the existing constructed social norms even when humans entirely withdraw observation.

Framing group alignment as a principal-agent problem simplifies misalignment in two places: information asymmetry (where principals only observe y_i but not θ_i and goal conflict. Full observation is costly and doesn’t scale with growing number of agents (Raubal et al., 2026; Kwa et al., 2025)).

The implication of this research is a cost-effective and time efficient way to ensure group alignment in AI agent mixed swarm deployment, after bespoke LLMs are trained.

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